

VR-based Equipment Training for Health Professionals

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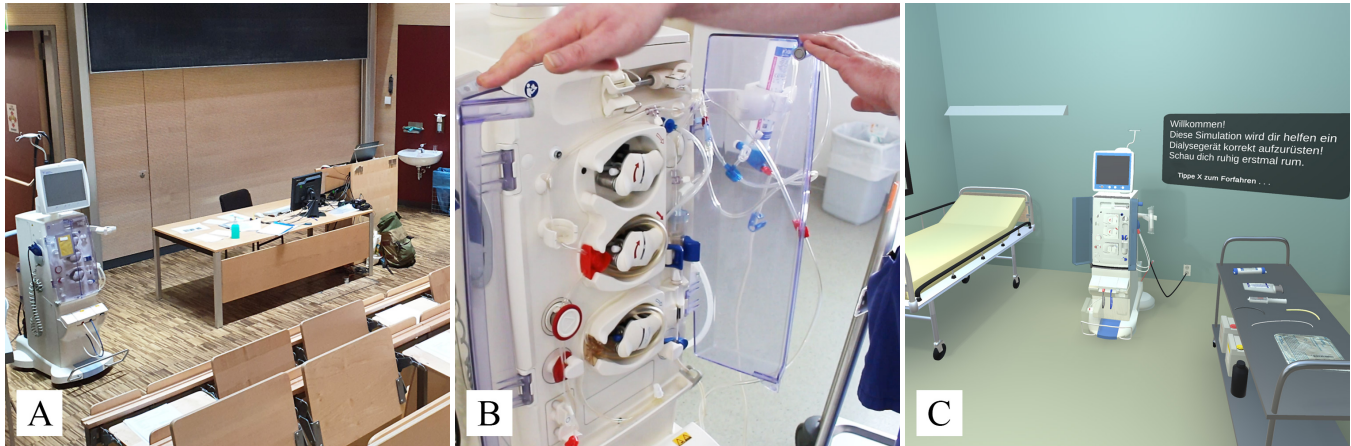


Figure 1: We evaluated three training methods for setting up a dialysis machine: *Group Training* (A), *Video Training* (B), *VR Training* (C)

ABSTRACT

It is common knowledge that health professionals' device training is a major problem, as the inadequate quality of device training puts patients at risk [2]. To find a way to counter this problem, we propose a virtual reality training simulation. The corresponding use case is exemplified by a priming procedure of a dialysis machine. This is achieved by users going through sequential interaction tasks in *Virtual Reality Training* using a head-mounted display. We evaluate this training method's potential within a user study, comparing it to traditional training methods, *Group Training*, and *Video Training*. Our findings demonstrate that *Virtual Reality Training* using a head-mounted display is a more effective and efficient method of learning how to prime a dialysis machine. Compared to other training methods, *Virtual Reality Training* takes longer on average, but the resulting learning effect is also higher. Furthermore, VR-training is more cost-effective than personal training and can

be repeatedly performed as it dispenses with the need for teaching professionals.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **Virtual reality**.

KEYWORDS

Virtual Reality, Medical Training, VR, Instruction Manual, Dialysis, User Studies, Equipment Training

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1 INTRODUCTION & RELATED WORK

A pandemic requires many new medical devices to be distributed within a short time. In order to operate these devices correctly, medical staff must be trained accordingly. Nowadays, such training is regulated so that usually an expert of the manufacturer carries out the training on-site, using a device with a group of trainees. However, studies and the experience of various experts from different countries have shown that this type of training is

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already insufficient under normal conditions [2]. There are several reasons for this: The training groups are too large and the training time too short. Also, the predetermined training dates are a problem, as this rarely occurs, and trainees are prevented from attending due to other priorities. Likewise, it is only possible to a limited extent to apply the learned content directly on the medical device. The reason is that the training only takes place on one device with one trainer, so it would be too time-consuming to let all participants train directly at the device. Since this directly affects at the expense of patient safety despite a certain tightening of regulations, we investigate an alternative training method. [4] Our method incorporates virtual reality (VR) because it does not cause the problems just mentioned, and it could reduce the costs and risks of the training [7]. In virtual reality, information can be conveyed more engagingly and understandably than it is possible in real life. Among other things, specific details can be visually highlighted and complex processes simulated. Furthermore, it is possible to adapt to the user's information transfer speed and repeat the desired contents as interaction steps. Similarly, *VR Training* is mainly independent of location, specific resources, and face-to-face contact. *VR Training* is gaining more and more interest: Werrlich et al. [3] compare paper, tablet, head-mounted display (HMD) based augmented reality, and in-situ projected instructions with 32 steps. A study conducted with 16 subjects demonstrates that assembling parts with the in-situ projected is significantly faster, and the error rate is the lowest. Funk et al. [11] differentiate paper-based training with AR training for engine assembly training, including 15 assembly steps. This study shows that AR training participants perform significantly fewer errors and take significantly more time. Winther et al. [12, 13] compare Pairwise Training, Video Training, and VR Training for the dosing pump's maintenance task. The study ($n = 36$) shows that *VR Training* is significantly lower in terms of the number of errors and completion time.

In this paper, we explore different training methods for priming a dialysis machine. The following three training methods (see Figure 1) are evaluated in a user study ($n = 83$): *VR Training*, *Video Training*, and the traditional training method of *Group Training* by an expert. Regarding these training methods, we investigate the following hypotheses:

- H1:** The learning success in *VR Training* is higher than in traditional *Group Training*.
- H2:** *VR Training* is more motivating than traditional *Group Training* and *Video Training*.

2 METHODS: DIALYSIS TRAINING

As a use case, we focus on hemodialysis's priming process, which is used to act as an artificial kidney to cleanse the blood in kidney failure cases. Various suppliers produce a device for this procedure. We chose the CorDiax 5008s¹ model by Fresenius Medical Care as it is a current device and one that is accessible to us. To define the correct training content for this machine, the manufacturer's official manual and a medical expert in the field of dialysis training

serve as input. A total of 60 relevant training steps are defined, which are integrated into all three training methods. Seventeen of these steps are informative, which defines specific details to the user, such as checking a connection or conveying theoretical content that explains the reason or consequences of a task. The remaining 43 steps, shown in Table 1, are tasks that the user must perform directly on the device. Therefore, they are performed by the *VR Training* users in a virtual environment.

2.1 Group Training

This traditional training type is used for training medical staff. Here, a group consisting of participants is trained simultaneously by an expert in one room. The expert demonstrates the specific content on the device and answers questions from the audience. Our training approach also occurs under the same conditions, where medical dialysis experts are demonstrating the training live on the device. Study participants are allowed to ask questions during the study time, just like participants in the real traditional training. Due to hygiene regulations, the training takes place in a lecture hall, as shown in Figure 1 (A), to ensure a minimum distance among the participants. However, the results should not be negatively affected by this, as the seats' pitch prevents students from blocking each other's view.

2.2 Video Training

For the training video's visual content, a healthcare professional is recorded while priming the dialysis machine. Care is taken to ensure that this person adheres to the defined training steps. For the auditory content, a professional speaker recorded these training steps, which were accompanied by light background music in the video. The training video is viewed by the study participants individually using an iPad. This is done under controlled conditions without a time limit. Figure 1 (B) shows a frame of the video. They are allowed to interact with the video in the following ways: rewind and fast-forward the video several times and adjust the volume individually.

2.3 VR Training

The application is being developed for the all-in-one VR glasses Oculus Quest², which does not require additional hardware, apart from the two accompanying wireless controllers, which provide a high degree of flexibility. This flexibility makes it possible, among other things, for medical staff to complete *VR Training* in any location they choose. For the dialysis machine and its accessories to be visible in the virtual environment, these components must first be measured and then digitally reconstructed using the 3D creation suite Blender³. Usually, the device is operated in a patient's room, which is also modeled and integrated into the virtual environment. This room positively influences the immersion as well as the mental performance of the user [8]. The software is implemented with the Unity⁴ development environment, taking care not to distract the user with complex interactions or graphic details. Interactions are crucial, positively impacting the user's engagement, attention, and

¹<https://www.freseniusmedicalcare.com/de/medizinisches-fachpersonal/haemodialyse/maschinen/5008-cordiax-5008s-cordiax/>

²<https://www.oculus.com/quest/>

³<https://www.blender.org/>

⁴<https://unity.com/>

Table 1: Sequence of steps that need to be carried out by the user for the dialysis machine's priming procedure. In training, these steps are explained in more detail to the user.

No.	Interaction step of the user	No.	Interaction step of the user
1	Insert dialyzer	23	Insert heparin syringe
2	Insert citrasate canister	24	Connect the heparin syringe to the arterial hose
3	Connect canister with hose	25	Place venous tubing system in the air
4	Insert sporotal	26	Place bubble catcher in level detector
5	Connect sporotal with tubing	27	Attach venous tubing system to upper right dialyzer port
6	Activate switch behind the machine	28	Connect pressure drain to pressure port
7	Operate switch on the screen	29	Insert the hose into the hose clamp
8	Select a treatment program	30	Attach the rinsing plug from the hose system
9	Open the unit doors	31	Take safeline out of the package and place it in the air
10	Open flap for acid connection	32	Place safeline fixation piece into the substitute pump
11	Place acid	33	Connect one end to the arterial tubing system
12	Connect the concentrate suction rod to the acid	34	First, place the other end at the bottom of the device
13	Close flap from the acid connection	35	Pull and turn the irrigation port counterclockwise
14	Open flap of dry bicarbonate concentrate	36	Attach the irrigation plug to the irrigation port
15	Insert dry bicarbonate concentrate	37	Connect the irrigation port
16	Close the flap of the dry bicarbonate concentrate	38	Turn and pull substitute port to attach substitute plug
17	Place arterial hose system in the air	39	Connect one end of the safeline to the substitute port
18	Insert fixation piece into the lower blood pump	40	Connect port
19	Insert the hose into the hose guide	41	Attach red drain hose to lower part of the filter
20	Attach arterial tubing system to dialyzer port	42	Attach blue inlet hose to the upper part of the filter
21	Pass the tubing under the door	43	Close the instrument doors
22	Lay down tubing		

concentration while learning [14]. Accordingly, the user sees two hands in the virtual environment, which follow the positions (6-degree-of-freedom) of the Oculus controllers. With these, users can perform three different interactions in training. The first interaction is pressing the x button on the left controller, which confirms the 17 informative steps. This is necessary since there is no interaction with objects that the system can use to recognize the successful completion of current steps. The second interaction involves touching objects, which happens when the virtual hand touches a virtual object, which in turn performs a particular action. As shown in Figure 2 (A), the third interaction is moving objects. Here the user has to press and hold the trigger on the controller while the virtual hand is in contact with a virtual object. It simulates the grabbing gesture by which objects can be moved in the virtual environment. If they are dropped in the air, they reappear at their original position. In contrast, objects that have reached their target position snap to their intended position and align themselves as intended. Since the training is a step by step training, the user must be taught the necessary interaction steps. Various methods are used for this purpose. The respective steps are displayed to the user as textual cues in the virtual environment (see Figure 2 (A, B, C)). Likewise, in Figure 1 (C), there are specific labels on the relevant components. Since auditory playback allows for better learning and less cognitive load than textual playback [1], the same voice instructions from the video training are used in addition to the textual cues, thereby guiding the user through the same sequential order of steps. After

completing a step, a beep sounds as confirmation, which positively influences the training [5]. Another interaction hint is the color highlighting of the objects and their target position, as shown in Figure 2 (A). Inserting the three sets of hoses from steps 17, 25, and 31 is challenging. A physically correct and appealing rendering of an interactive hose requires specific computational capabilities limited in all-in-one VR glasses like the Oculus Quest. Therefore, the hose set is not treated as real but first placed in the air next to the device (see Figure 2 (B)). It is divided into several segments, enabling the user to place the segments one after the other in the correct position (see Figure 2 (C)). Before the subjects start the *VR Training*, they have to complete a tutorial. This ensures that they are not overwhelmed or distracted by the virtual experience, the novelty of this technology, or the interaction techniques. This tutorial is a VR scene in which users can familiarize themselves with a virtual environment and test the interaction techniques. The tutorial and the *VR Training* which follow take place without time limitations under controlled conditions.

3 EVALUATION

We adapted a between-subjects design to compare the three different training methods, defined as the independent variables. To identify and correct potential problems early, we conducted a preliminary study ($n = 5$, male = 4, and female = 1) for the *VR Training*.

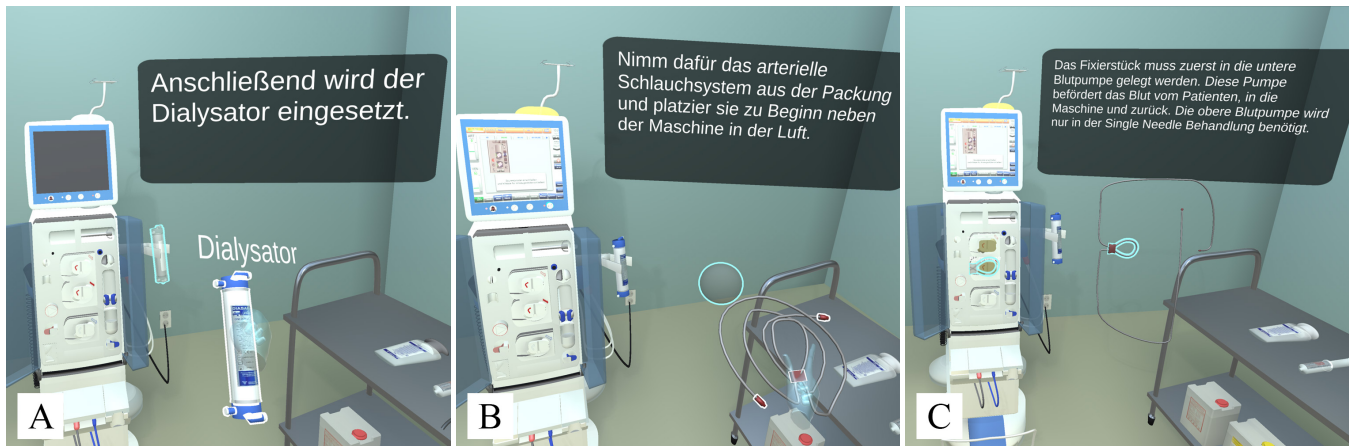


Figure 2: (A) shows moving and placing virtual objects with visual hints, (B) shows an easier interaction technique insertion of a hose set, (C) shows a hose set, which is disassembled into individual components next to the device

This preliminary study's most interesting result was that four subjects complained about light nausea and vertigo in a subsequent interview, which are cybersickness symptoms [6]. To counteract this, we have placed all the components needed for priming (acid canister, citrasate canister, and the sporotal bottle), which were initially placed under the table shown in Figure 1 (C), on the table to reduce movements that require the user to bend down.

3.1 Apparatus

The CorDiax 5008s dialysis machine was used for the *Group Training* along with the required accessories. Subjects of the *Video Training* viewed the video on an iPad 11 inch with a resolution of 2388×1669 pixels via the iOS Photos application. The training video lasts 05:45 minutes and has a resolution of 1920×1080 pixels at 25 fps. The 1st generation Oculus Quest VR glasses were used for the *VR Training*. These have a refresh rate of 72 Hz at a resolution of 1440×1600 pixels per eye. Unity version 2019.3.10f1 was used as the development environment. The required 3D models were constructed with Blender version 2.80.

3.2 Procedure

The experiment was divided into the following four phases, with the first three phases taking place on a clinic's premises. In the first phase, subjects completed a questionnaire with metadata as well as their previous experiences. In the second phase, the subjects finished one of the three training methods. The *Group Training* duration ranged from 7:44 to 9:14 minutes ($M = 8.4$, $SD = 0.56$). In total, there were five sessions in which between two and eight subjects participated. The *Video Training* completed by subjects took between 5:45 to 14:06 minutes ($M = 7.1$, $SD = 2.68$), the *VR Training* between 5:39 to 18:37 minutes ($M = 10.1$, $SD = 2.68$) and the *VR Tutorial* between 01:12 to 06:20 minutes ($M = 3.3$, $SD = 1.16$). Immediately after the training, the subjects completed a questionnaire consisting of various items. The learning effect was assessed using the FIRE-B questionnaire [9], and the content's structure and preparation were evaluated using the validated MFE-ZcL questionnaire [10]. Furthermore, a general evaluation of the training, as well

as its content, was questioned. The fourth phase served to check the learning success, which we recorded with an online test that the subjects received by e-mail precisely seven days after completing the training. This time offset was used to obtain more meaningful results concerning long-term learning success. The test consisted of 16 questions, single-choice or matching questions, depending on the content. This test queried at which positions specific components are placed or connected, as well as their designation. It also checked the order in which specific steps are performed. Since the online test did not take place under laboratory conditions, the subjects were instructed after training and before starting the online test to refrain from using additional aids that could influence the test result. Completion of the online questionnaire was limited to 25 minutes.

3.3 Limitations

Since this use case is a medical product, we explicitly chose subjects with an active medical background who had not yet undergone dialysis training. Therefore, only individuals from the following professional groups participated in this study: Doctors, nurses, and medical students/doctoral fellows. To recruit these participants, we used a university mailing list, flyers, and social networks. Similarly, we emphasized equal distribution to ensure that they were mainly distributed equally according to their age, gender, and occupational group.

3.4 Participants

A total of 83 subjects (male = 28 and female = 55) participated in this study. The subjects ranged in age from 18 to 59 years ($M = 30.4$, $SD = 9.64$). They were divided among the respective training groups, as indicated in Table 2. Six subjects of the *VR Training* had already gained first experiences with virtual reality.

4 RESULTS

In the following, we present the results of the post-test questionnaire. To test for significant differences between different training

Table 2: Distribution of participants to the respective training groups.

	n	Age		Gender		Profession		
		M	SD	Male	Female	Doctor	Nurse	med. Student / PhD
Group Training	28	28.6	7.41	8	20	7	3	18
Video Training	27	28.7	9.51	9	18	5	9	13
VR Training	28	33.9	11.02	11	17	8	11	9

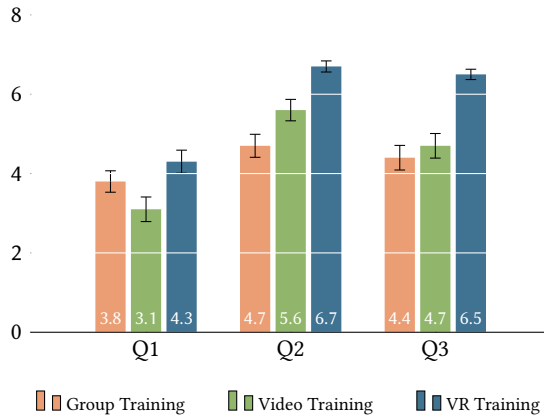


Figure 3: Plot with the results of the post-test questionnaires assessments per training method. It shows scores in the three categories. Error bars indicate standard error. Q1 to Q3 were rated on a 7-point Likert scale [strongly disagree, strongly agree].

methods, we applied the non-parametric Kruskal-Wallis test ($\alpha = 0.05$). For the subsequent posthoc tests, the Mann-Whitney test with Bonferroni correction was used, and effect sizes (r) are given as Cohen's d . Figure 3 shows the results grouped by the three training methods and the respective questions (Q), where Q1 to Q3 were rated on a 7-point Likert scale [strongly disagree, strongly agree]. Figure 4 presents the results of the questions Q4 and Q5, which were rated on a 6-point Likert scale [very good, not sufficient].

Q1: "I feel able to implement what I have learned in the exercise independently." For evaluating the learning effect, the VR Training received a significantly better rating compared to the Video Training ($\chi^2 = 7.390$, $p = 0.02$; Posthoc: Video Training - VR Training $p = 0.02$, $r = 0.73$).

Q2: "I think this exercise was clearly structured." The posthoc test revealed significant differences between all training methods, with VR Training performing best ($\chi^2 = 28.273$, $p < 0.01$; Posthoc: Group Training - Video Training $p = 0.03$, $r = 0.62$ — Group Training - VR Training $p < 0.01$, $r = 1.64$ — Video Training - VR Training $p < 0.01$, $r = 0.95$).

Q3: "I think the exercise topic was interestingly prepared." In terms of interesting preparation, VR Training was rated significantly more interesting than the other two training methods ($\chi^2 = 30.809$, p

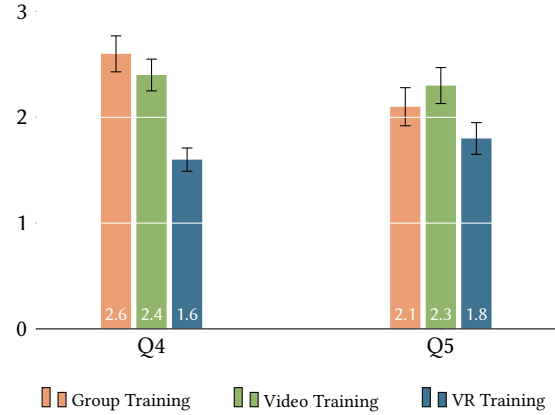


Figure 4: Results from the questions Q4 and Q5 were rated on a 6-point Likert scale [very good, not sufficient]. Lower values correspond to a higher agreement. Error bars indicate standard error.

$p < 0.01$; Posthoc: Group Training - VR Training $p < 0.01$, $r = 1.69$ — Video Training - VR Training $p < 0.01$, $r = 1.49$).

Q4: "Which grade would you give the exercise overall?" In general, VR Training was rated significantly better than the other two training methods ($\chi^2 = 21.089$, $p < 0.01$; Posthoc: Group Training - VR Training $p < 0.01$, $r = 1.32$ — Video Training - VR Training $p < 0.01$, $r = 1.11$).

Q5: "What grade would you give the quality of the content?" There were no significant differences in the quality of the content. The VR Training took significantly more time compared to Group Training and Video Training ($\chi^2 = 28.815$, $p < 0.01$; Posthoc: Group Training - VR Training $p = 0.01$, $r = 0.89$ — Video Training - VR Training $p < 0.01$, $r = 1.14$).

In the online test conducted to measure the learning success, 78 of the original 83 subjects participated. On average, 43.23 % of the questions were answered correctly in the Group Training ($n = 24$), 49.52 % in the Video Training ($n = 26$) and 50.67 % in the VR Training ($n = 28$).

5 DISCUSSION

The experiment results support the hypothesis that it is more motivating for participants to pass through the VR Training than the

other two training methods (H2). The users' motivation supports this hypothesis by the interesting preparation (Q3) of the *VR Training*. Likewise, the more obvious *VR Training* structure (Q2) does not negatively influence the motivation during the training. The training duration, which was significantly longer in the *VR Training* than in the *Group Training* and *Video Training*, reflects previous work [12, 13] results, which could also indicate increased motivation. However, the time differences are as expected since the *Group Training* follows a strict schedule, and the time varies mainly due to questions asked by the participants. On the other hand, compared to the other two training methods, *VR Training* offers the advantage that users can view significantly more details from different perspectives and interact with the system themselves, which impacts the training duration. Furthermore, the results indicate that *VR Training*'s learning success is higher than traditional *Group Training* (H1). This hypothesis is supported by the fact that after the training, significantly more participants felt able to independently implement what they had learned (Q1), which in turn was confirmed by the test results. Despite the similar training methods, the results of learning success differ from those of Winther et al. [12, 13]. This is because they used pairwise training and not *Group Training* as we did. Likewise, the methods used to test learning success differ, as does the elapsed time to test learning success. Given these results, it is very tempting to assume that virtual reality-based training should be used more often to learn medical skills and medical devices. It saves human resources, is very cost-effective after the application has been developed, and can be easily adapted to new versions of medical devices. Given the limited resources available for developing the VR-based application for this study, we are very optimistic that with the inclusion of more interaction, gamification approaches, an avatar guiding the learning steps, and in-depth background teaching modules, the learning effects might be even higher. It can not be mentioned enough that it should always be the goal of teachers to make teaching not only effective but also enjoyable. It seems that VR-based medical training might be one step in the right direction to achieve that goal.

6 CONCLUSION

In this paper, different training methods for priming hemodialysis are presented and compared. These methods are traditional *Group Training*, *Video Training*, and *VR Training*. To measure the learning success of these methods, they are compared in a study with 82 participants. These study results indicate that the implemented *VR Training* is more effective than the traditional *Group Training* in terms of learning success and more efficient due to the given flexibility (e.g., physical availability of the device or scheduling). Thus, *VR Training* has the potential to have a positive impact on patient safety.

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